

Project Plan

Due 10/9/2025

1. Group:

- a. Samir Husen – U01109666
- b. Aayush Dhakal – U01117654

2. Project title: Breast Cancer Classification

3. Project Dataset:

- a. **Link to dataset:** <https://www.kaggle.com/datasets/nancyalaswad90/breast-cancer-dataset>
- b. **Target (Supervised):** diagnosis — (M = Malignant, B = Benign)
- c. **Features:**

Id, diagnosis, radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, compactness_mean, concavity_mean, symmetry_mean, fractal_dimension_mean.

d. Brief description:

The dataset contains 569 samples and 32 features including the target variable. Each sample represents a breast mass with features computed from a digitized image of a fine needle extract (FNA) of a breast mass. The features characterize various attributes of the visible cell nucleus in the digital image.

4. Timeline

Date	Student	Task
Oct 10–13, 2025	Samir Husen	Data Preprocessing (handle missing values, encode target, normalize)
Oct 14–17, 2025	Aayush Dhakal	Exploratory Data Analysis (visualizations, correlations, outliers)

Oct 18–21, 2025	Samir Husen	Feature Selection (correlation thresholding, RFE, feature importance)
Oct 22–25, 2025	Aayush Dhakal	Data Scaling (StandardScaler, MinMaxScaler)
Oct 26–29, 2025	Aayush Dhakal	Model Implementation: Logistic Regression, KNN, Decision Tree, Random Forest
Oct 30–Nov 3, 2025	Samir Husen	Model Implementation: SVM, XGBoost, Neural Network
Nov 4–8, 2025	Aayush Dhakal	Hyperparameter Tuning
Nov 9–14, 2025	Samir Husen	Model Evaluation & Comparison (Accuracy, Precision, Recall, F1)
Nov 15–22, 2025	Samir & Aayush	Report Writing and Visualization Integration
Dec 7, 2025	Samir & Aayush	Final Presentation + Recording Submission

5. Planned Algorithms

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine
- XGBoost / Gradient Boosting
- Neural Network

6. Outcome

The final deliverable will include:

- A report covering the entire ML pipeline
- Performance comparison table
- Visualizations for each phase
- Presentation + recorded demo